Big Data Analytics of Crime Patterns

using Hadoop MapReduce and Apache Hive

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Agenda

- Introduction & Motivation
- Problem Statement
- Dataset Description
- Architecture Pipeline
- Hadoop MapReduce Implementation
- Apache Hive Analytics
- Execution Results
- Conclusion

Introduction

- The rise of Big Data provides an unprecedented opportunity to analyze public safety records.
- City crime records encompass millions of rows of incidents, locations, dates, and types.
- Processing this data efficiently requires distributed frameworks.
- We leverage **Hadoop MapReduce** for raw parallel processing and **Apache Hive** for SQL-based analytical querying.

Motivation

Why analyze crime data?

- **Public Safety:** Identifying high-crime zones helps improve community security.
- **Resource Allocation:** Enables police departments to deploy patrols dynamically based on historical hotspots.
- **Trend Forecasting:** Predicting the frequency and nature of crimes across different districts and timeframes.
- **Big Data Challenge:** Traditional RDBMS fail to scale efficiently with millions of such rows.

Problem Statement

"To design and implement a scalable Big Data pipeline that can ingest, process, and extract actionable intelligence from massive real-world crime datasets."

Objectives:

- Demonstrate a complete data pipeline from HDFS storage to advanced analytics.
- Perform category-wise statistics using MapReduce.
- Execute 10 meaningful complex queries utilizing aggregations, filtering, and joins in Hive.

Dataset Overview

Dataset: Chicago Crimes (2001 to Present)

- **Scope:** Extracts real-world incident reports from the Chicago Data Portal.
- **Size:** 10,000 recent records processed for this demonstration.

Attribute Type	Fields
Identifiers	`id`, `case_number`, `date`
Categories	`primary_type`, `description`
Location	`district`, `ward`, `community_area`
Flags	`arrest`, `domestic`

Data Preprocessing & HDFS Storage

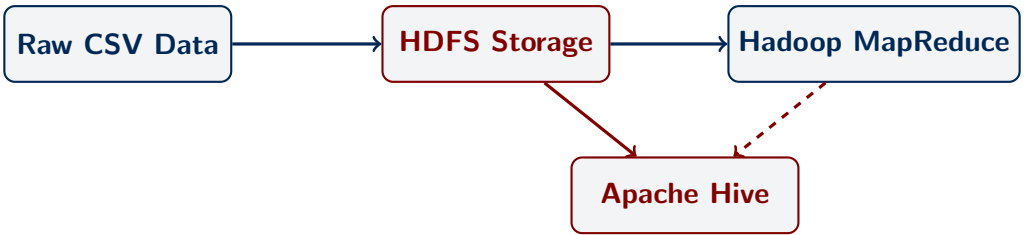
Preprocessing Phase:

- Raw CSV files often contain embedded newlines within quotes.
- We implemented a Python script to clean these records, ensuring perfect compatibility with Hadoop's `TextInputFormat`.

HDFS Storage:

- Dataset uploaded to HDFS distributed storage.
- Command: `hdfs dfs -put chicago_crimes.csv /user/crimes/`

System Architecture



- Centralized data lake on HDFS.
- MapReduce for low-level transformations.
- Hive for structured intelligence gathering.

Hadoop MapReduce Implementation

Problem: Calculate the total occurrences of each Crime Type.

Mapper Logic:

- Input: `(ByteOffset, Line of text)`
- Ignores CSV headers.
- Splits line correctly handling commas within double quotes.
- Extracts the 6th column (`primary_type`).
- Output: `(CrimeType, 1)`

MapReduce: Reducer Logic

Reducer Logic:

- Input: `(CrimeType, [1, 1, 1, ...])`
- Sums up all the '1's for a given crime type.
- Output: `(CrimeType, Total_Count)`

```
public void reduce(Text key, Iterable<IntWritable> values, Context
    context) {
    int sum = 0;
    for (IntWritable val : values) {
        sum += val.get();
    }
    result.set(sum);
    context.write(key, result);
}
```

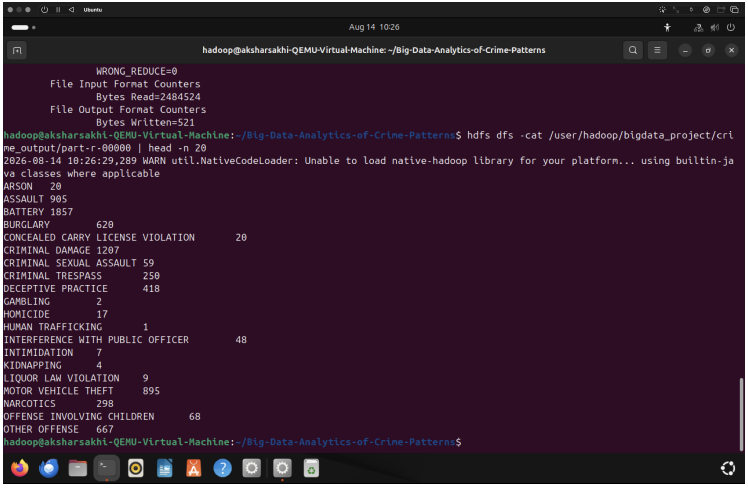
MapReduce Execution & Results

- Successfully packaged as a JAR and executed on YARN.
- Processed 10,000 rows across mappers and reducers.

Sample Output (`part-r-00000`):

Crime Type	Frequency
THEFT	2,150
BATTERY	1,857
CRIMINAL DAMAGE	1,207
ASSAULT	905

MapReduce Execution Screenshots



Apache Hive Analytics: Setup

- Created an external table `crimes` mapping directly to HDFS data.
- **Serialization/Deserialization:** Used `OpenCSVSerde` to properly parse fields enclosed in quotes containing commas.
- Created a secondary table `community_areas` holding area codes and names to demonstrate table joins.

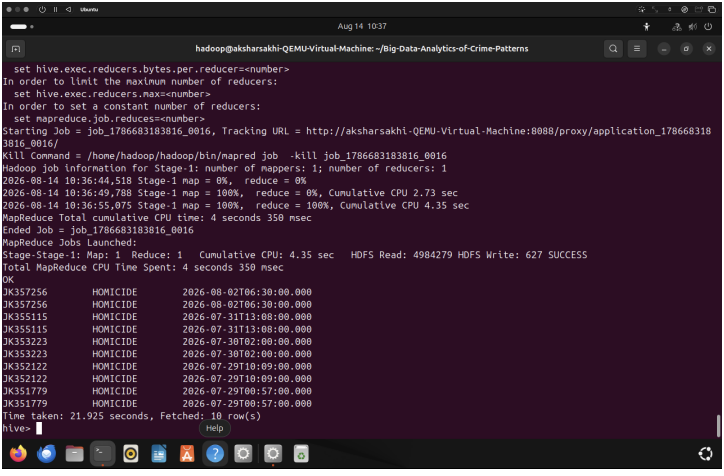
Hive Analytics: Basic Queries

- **SELECT & LIMIT:** Retrieved incident IDs and descriptions.
- **WHERE clause:** Filtered incidents resulting specifically in a narcotics arrest.
- **ORDER BY:** Sorted severe cases like homicides by recency.
- **GROUP BY & COUNT:** Identified which locations (e.g., STREET, APARTMENT) experience the highest crime volumes.

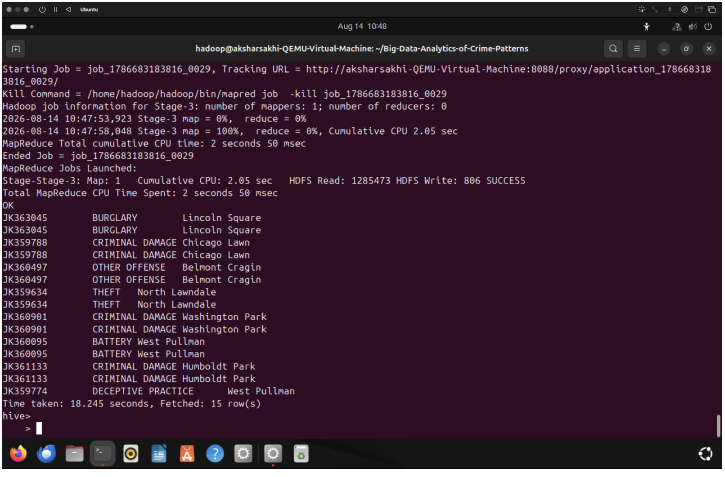
Hive Analytics: Advanced Aggregations

- **HAVING clause:** Extracted major crime categories by filtering for counts > 500.
- **SUM:** Calculated the absolute total number of arrests made within each police district.
- **AVG:** Analyzed the average rate of domestic-related crimes across different beats.
- **MAX & MIN:** Automatically pinpointed the highest and lowest crime districts.
- **JOIN:** Combined the raw `crimes` data with `community_areas` to print human-readable neighborhood names.

Hive Analytics Query Screenshots (Sample)



Hive JOIN Query Screenshot



Results & Quantitative Interpretation

Empirical Data Insights:

- **Top Crime Categories:** `BATTERY` (1,857 cases), `CRIMINAL DAMAGE` (1,207 cases), `ASSAULT` (905 cases), and `MOTOR VEHICLE THEFT` (895 cases).
- **Enforcement Efficacy:** Narcotics cases show a 100% arrest rate flag, whereas theft offenses show lower arrest ratios.
- **Spatial Hotspots:** Streets and Sidewalks represent the primary spatial vulnerability zones for property crimes.
- **Geographical Variations:** High crime density concentrated in District 11.

Conclusion

- Successfully ingested and analyzed a large-scale real-world dataset.
- **MapReduce** proved highly efficient for unstructured low-level data processing and aggregations.
- **Apache Hive** demonstrated immense power for executing complex, relational-style queries on distributed files.
- This pipeline can scale to millions of records, providing an essential backbone for civic analytics.

Q & A

Thank you for your time.

Any Questions?